Directed numbers on the number plane

1

a $(6, 7)$ **b** $(3, 9)$ **c** $(4, 4)$ **d** $(9, 5)$

e $(-4, 3)$ **f** $(-9, 6)$ **g** $(-2, 7)$ **h** $(-7, 8)$

i $(-2, -8)$ **j** $(-6, -7)$ **k** $(-7, -2)$ **l** $(-5, -9)$

m $(3, -2)$ **n** $(4, -7)$ **o** $(8, -6)$ **p** $(7, -1)$

2

a $(1.5, 3)$ **b** $(-1.5, 0)$ **c** $(-2.5, -3.5)$ **d** $(3.5, -3.5)$

3

a **i** 0 **ii** 0 **b** **i** 0 **ii** 0

4 y-axis

5 $a = 0$

6

a Q **b** S **c** P **d** R

7

a $A\,(3, 0)$ **b** $B\,(-2.5, 0)$ **c** x-axis

8

a 1st quadrant **b** 2nd quadrant **c** 4th quadrant **d** 3rd quadrant

9

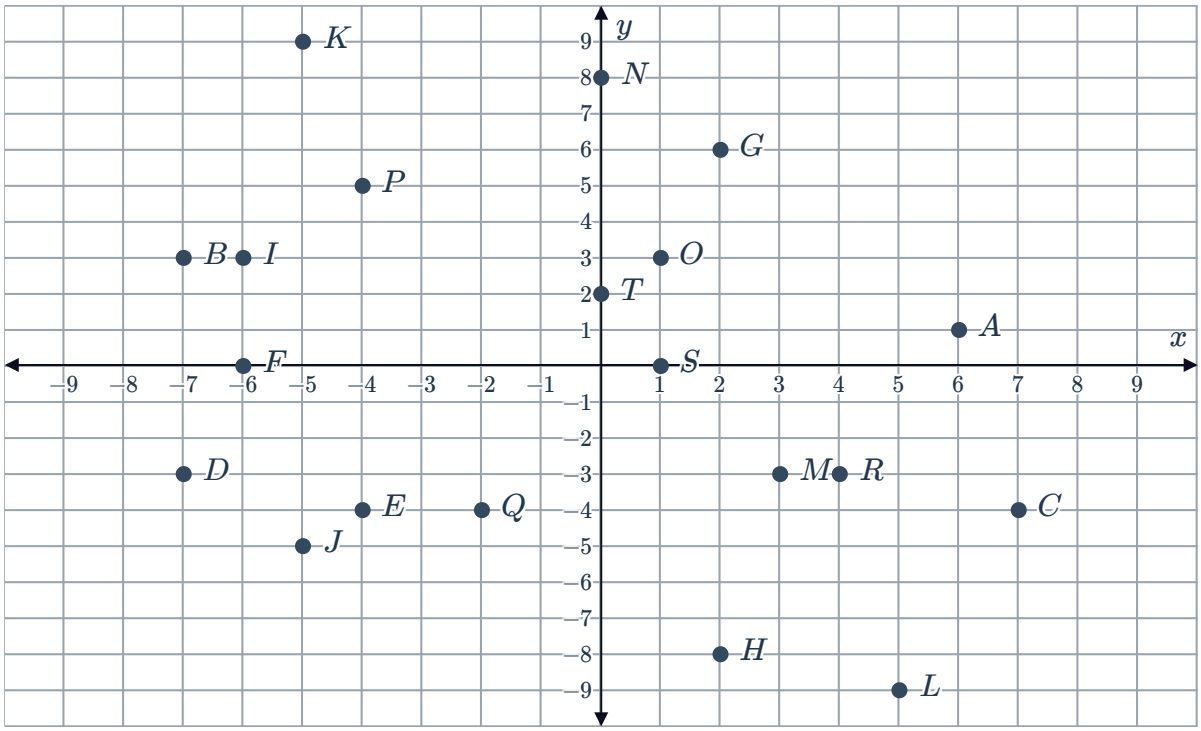
a 3rd quadrant, 4th quadrant **b** 2nd quadrant, 3rd quadrant

c 1st quadrant, 3rd quadrant **d** 2nd quadrant

e 3rd quadrant

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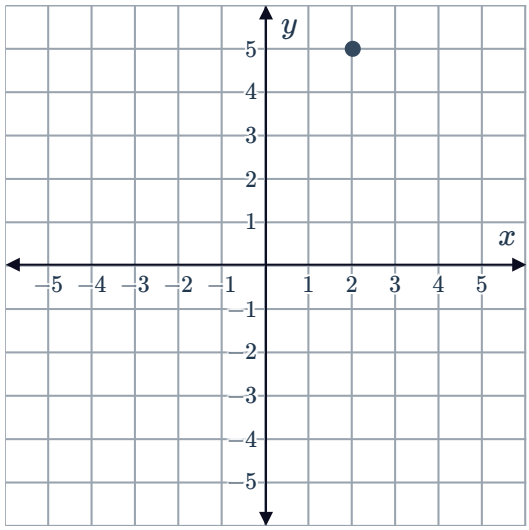
10



11

a

i

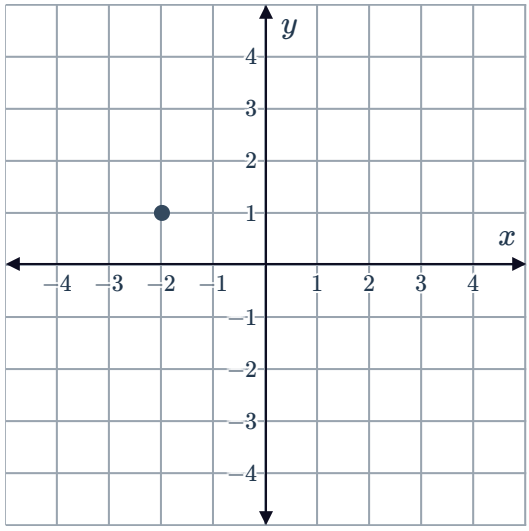


ii 1st quadrant

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b

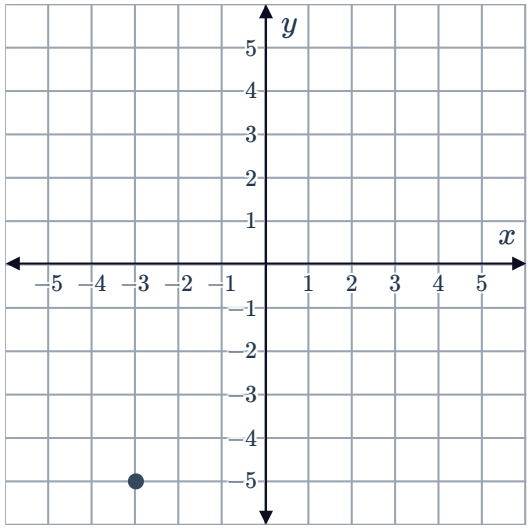
i



ii 2nd quadrant

c

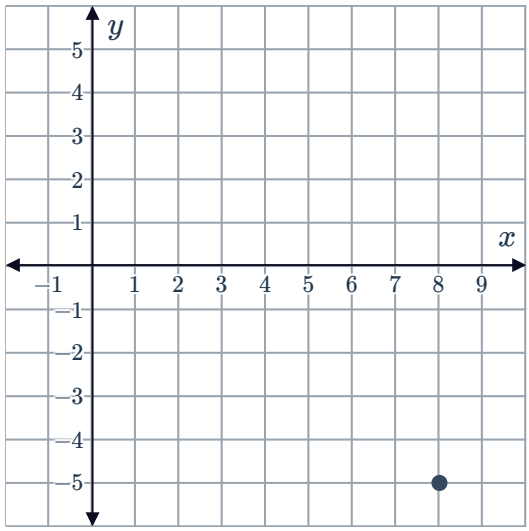
i



ii 3rd quadrant

d

i

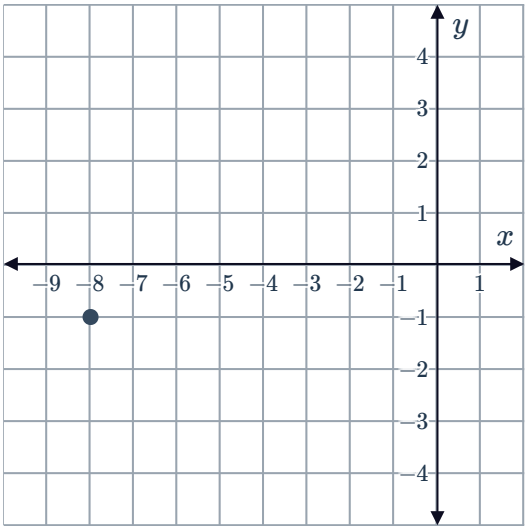


ii 4th quadrant

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e

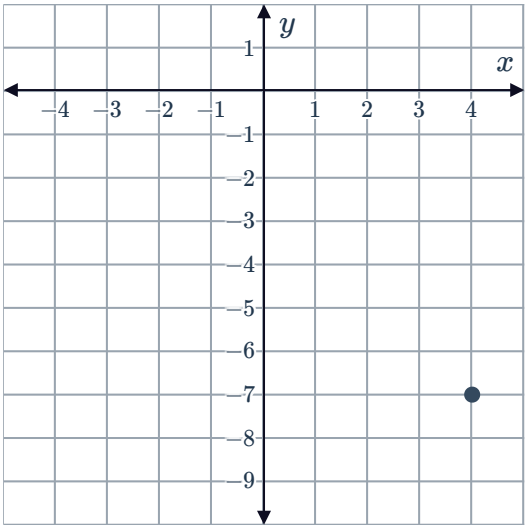
i



ii 3rd quadrant

f

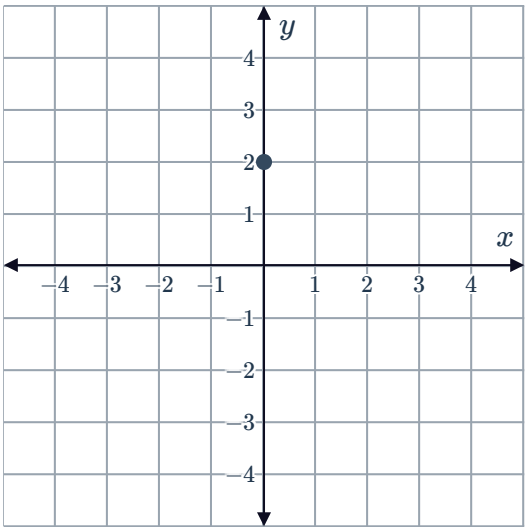
i



ii 4th quadrant

g

i

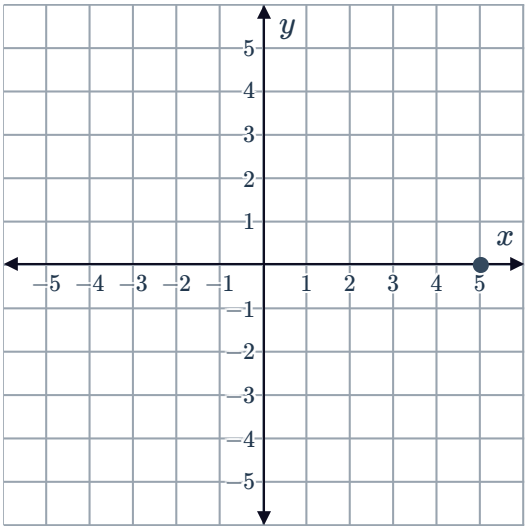


ii No quadrant

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h

i

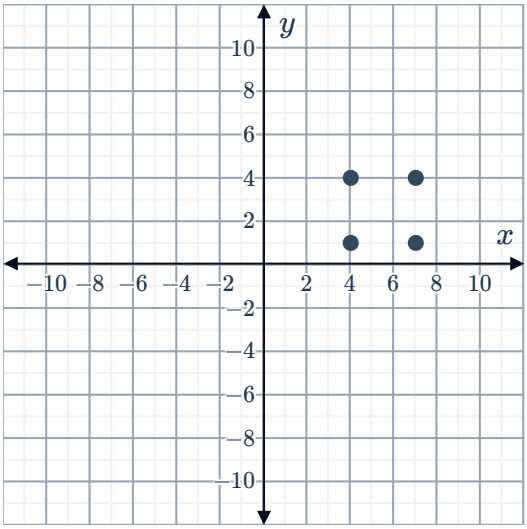


ii No quadrant

12

a

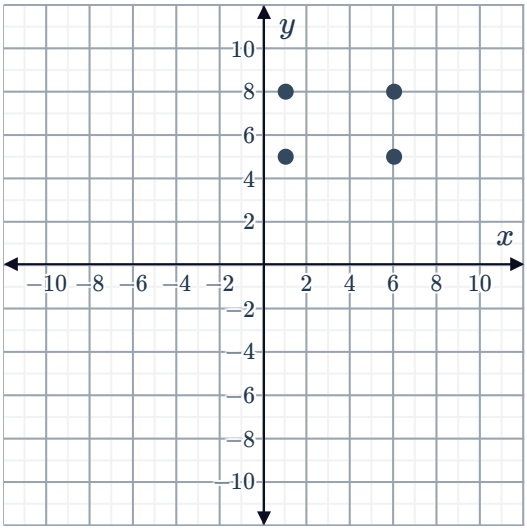
i



ii Square

b

i

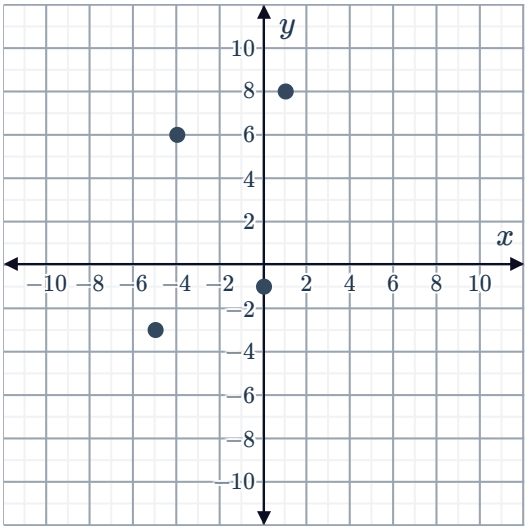


ii Rectangle

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c

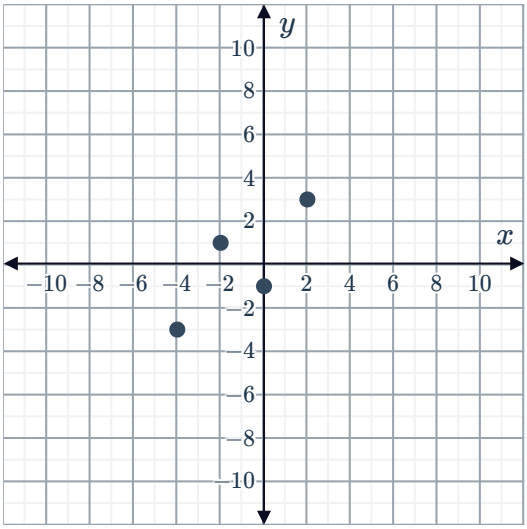
i



ii Parallelogram

d

i



ii Rhombus

13

a $A\,(5,9)\,, B\,(5,-3)\,, C\,(-2,-3)\,, D\,(-2,9)$

b $T\,(8,6)\,, U\,(8,-9)\,, V\,(-4,-9)$

c $A\,(6,2)\,, B\,(-9,3)\,, C\,(-9,-7)\,, D\,(6,-8)$

d $S\,(8,9)\,, T\,(-5,4)\,, U\,(-7,-3)\,, V\,(6,-6)$

14

a Move 2 units to the right, and then 4 units down.

b Move 9 units to the right, and then 14 units up.

15

a $(0,-9)$

b $(-3.5,0)$

c $(-7,6)$

d $(6,-2)$

e $(4,3)$

f $(-2,1)$

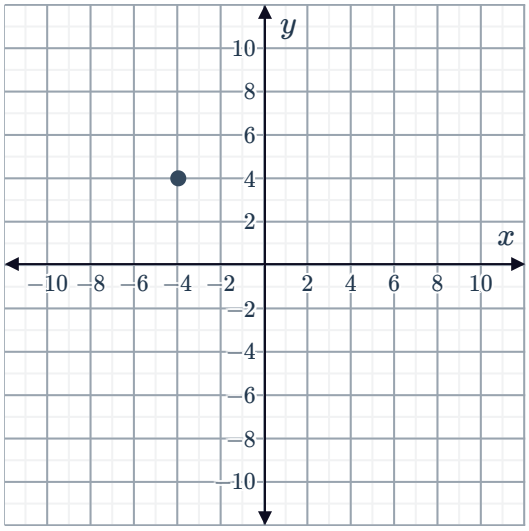
16 **B**

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17 1

18

a



b $(-4, 4)$

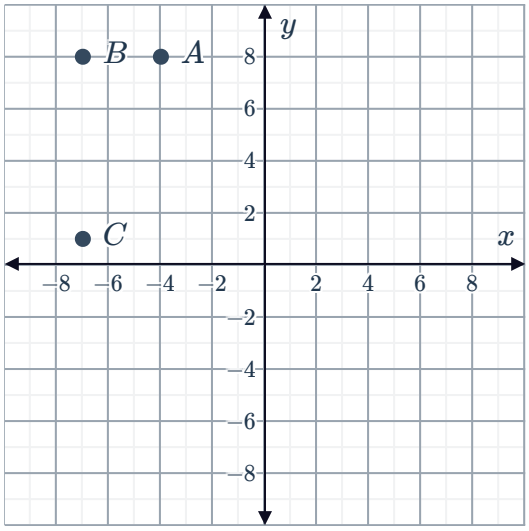
Distance between two points

19

a 3 units **b** 8 units **c** 4 units **d** 9 units

20

a



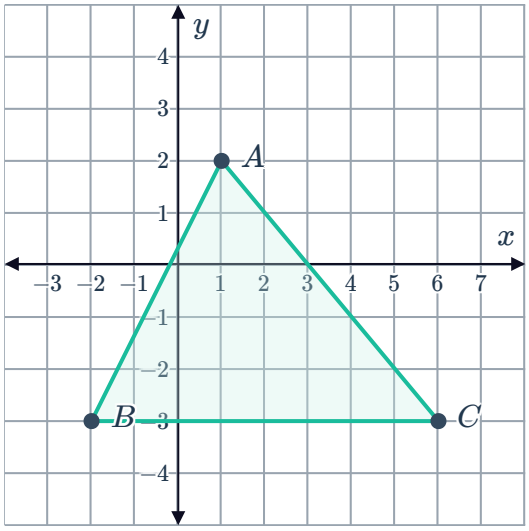
b 3 units

c 7 units

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21

a



b 5 units

c 8 units

d 20 units^2

Geometric applications

22 $(-2, 2)$

23 $(-2, -1)$

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